Hybrid arithmetic holonomy and twenty-two individual p-adic zeta values

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Abstract

We develop a hybrid finite-place refinement of arithmetic holonomy for the Eisenstein–Eichler connection on the genus-zero modular curves $X_0(p)$, $p \in \{2, 3, 5, 7\}$. The denominator staircase of the Eichler primitive cannot be realized by globally defined scalar solution coordinates: the nonzero modular weights create a genuine torus obstruction. We instead use two different local mechanisms. At auxiliary primes below a cutoff, the Hasse invariant algebraizes the unipotent comparison coordinate modulo the prime and a genuine-source rank–nullity argument produces one determinant-level saving on a large source kernel. Above the cutoff, we keep the calculation on the algebraic de Rham frame torsor; exact divided-Frobenius relations for the raw Eichler rows and a coefficient-ring Taylor no-backflow lemma recover the Calegari–Dimitrov–Tang prime window without scalarizing the packet. The resulting arithmetic cost is

$$\tau_{p,s}(\xi) = \frac{2}{(s+1)^2}\left(S_{p,s}(\xi) + sI_\xi^{s+1}(\xi)\right),$$

where $S_{p,s}$ is the small-prime Hasse-codimension cost and I_ξ^{s+1} is the published prime-window function. Combining this with Calegari's overconvergent Eisenstein family, Buzzard's analytic continuation, an exact modular Jensen formula, and Bost's slopes inequality proves the irrationality of at least twenty-two individual Kubota–Leopoldt values:

$$\zeta_2(s) \quad (3 \le s \le 29,\ s \text{ odd}), \qquad \zeta_3(s) \quad (3 \le s \le 11,\ s \text{ odd}),$$

$$\zeta_5(3), \quad \zeta_5(5), \quad \zeta_7(3).$$

Every numerical inequality is supplied with an independent rational fixed-point interval certificate. No opposite Hodge filtration, rational Tate scalarization, or rowwise transport of literal n^e denominators is used.

Audit status. The proof isolates five published interfaces: Calegari's overconvergent Eisenstein family, Buzzard's finite-slope continuation theorem, Katz's Hasse invariant formalism, the de Rham frame-torsor/Gauss–Manin construction of Graham–Pilloni–Rodrigues Jacinto, and the Bost–Charles/CDT slopes and prime-window estimates. The canonical BGG lift, single-action deepest-row projection, genuine-source Hasse kernel, hybrid local lattices, full-product pole-grade Cartier argument, collision calculation, and all numerical applications are proved in the manuscript. Because the conclusions include individual irrationality statements beyond the classical cases, independent specialist review remains appropriate before publication.

Contents

1 Introduction and main results

For an odd integer $s \geq 3$, let $\zeta_p(s)$ denote the Kubota–Leopoldt p-adic zeta value in the normalization used by Calegari. Calegari proved the irrationality of $\zeta_2(3)$ and $\zeta_3(3)$ in [4]; the irrationality of $\zeta_2(5)$ was later obtained through arithmetic holonomy and independently by an Apéry-type construction. General results also give many irrational values or finite alternatives, but usually do not identify a large fixed list of individual values; see, for example, [11, 12, 13]. We do not attempt a complete priority analysis here. Our theorem is the following fixed list.

Theorem 1.1 (Twenty-two individual values). *The following p-adic numbers are irrational:*

$$\begin{aligned} &\zeta_2(s) && (s = 3, 5, \ldots, 29), \\ &\zeta_3(s) && (s = 3, 5, \ldots, 11), \\ &\zeta_5(3),\ \zeta_5(5), && \zeta_7(3). \end{aligned}$$

The proof begins with the negative-weight Eisenstein series

$$K_{p,s}(q) = \frac{\zeta_p(s)}{2} + \sum_{n\geq 1} \sigma_{-s}^{(p)}(n) q^n, \qquad \sigma_{-s}^{(p)}(n) = \sum_{\substack{d|n \\ p\nmid d}} d^{-s}.$$

Under the contrary assumption $\zeta_p(s) \in \mathbf{Q}$, this is an arithmetic horizontal leaf of a rank-$(s+1)$ Eisenstein–Eichler connection. Its raw Euler derivatives have exact integration depths

$$0, s, s-1, \ldots, 1,$$

but the positive-depth rows have different modular weights. The diagonal torus part of a Hodge-frame change therefore cannot be represented by a bounded rational matrix in the Hauptmodul. A scalar filtered realization is not merely unavailable; it is incompatible with the modular transformation law.

The replacement is hybrid. At small auxiliary primes, the Hasse invariant makes the normalized unipotent comparison coordinate rational modulo the prime. A rank–nullity argument on the genuine global source then produces a kernel of the required codimension, allowing one reciprocal prime to be saved on

$$(s-1)\left(D - \frac{p+1}{12}\ell\right) + O_{p,s}(1)$$

source directions. At large auxiliary primes, the scalar pairing is computed upstairs on the de Rham frame torsor. The fibre coordinates remain coefficients, while only the raw scalar Eichler rows undergo the exact Dwork substitution. Residue classes modulo ℓ prevent Taylor backflow over any coefficient algebra and recover a degree-$D + O(1)$ prime window.

For $m = s+1$ define

$$K_m(\xi) = \left\lfloor \frac{m-1}{\xi} \right\rfloor$$

and

$$I_\xi^m(\xi) = \left(m - \frac{1}{2}\right) H_{K_m(\xi)} - K_m(\xi)\xi + \frac{(m - (K_m(\xi)+1)\xi)_+^2}{2(K_m(\xi)+1)}. \tag{1}$$

Put

$$c_p = \frac{p+1}{12}, \qquad J_p(\xi) = \int_0^\xi (1 - c_p x)_+ \, \mathrm{d}x$$

and

$$S_{p,s}(\xi) = s^2\xi - (s-1)J_p(\xi). \tag{2}$$

Thus, when $c_p\xi < 1$,

$$S_{p,s}(\xi) = s^2\xi - (s-1)\left(\xi - \frac{p+1}{24}\xi^2\right).$$

The main arithmetic statement is as follows.

Theorem 1.2 (Hybrid arithmetic holonomy)**.** *Fix $p \in \{2,3,5,7\}$ and an odd $s \geq 3$. Under the assumption $\zeta_p(s) \in \mathbf{Q}$, let ϕ be admissible adelic continuation templates for the rank-$m = s+1$ Eisenstein–Eichler leaf, with derivative width $\Lambda(\phi)$ and Bost–Charles energy $\mathscr{H}(\phi)$. For every $1 < \xi < m$ one has*

$$m\left(\Lambda(\phi) - \tau_{p,s}(\xi)\right) \leq \mathscr{H}(\phi),$$

where

$$\boxed{\tau_{p,s}(\xi) = \frac{2}{(s+1)^2}\left(S_{p,s}(\xi) + sI_\xi^{s+1}(\xi)\right).} \tag{3}$$

For the modular templates used below,

$$\Lambda_p(Y) = \frac{12\log p}{p-1} - 2\pi Y$$

and

$$\mathscr{H}_p(Y) = \Lambda_p(Y) + C_p(Y),$$

where

$$C_p(Y) = 4\pi \sum_{\substack{c\geq 1\\ p|c\\ cY<1}} \frac{\varphi(c)}{c^2}\left(\arccos(cY) - cY\sqrt{1-c^2Y^2}\right). \tag{4}$$

Consequently rationality would force

$$\mathfrak{M}_{p,s}(\xi, Y) := s\Lambda_p(Y) - (s+1)\tau_{p,s}(\xi) - C_p(Y) \leq 0. \tag{5}$$

For the twenty-two pairs in Theorem 1.1, exact rational witnesses ξ, Y make the left side strictly positive. The smallest independently certified lower bound is

$$\mathfrak{M}_{5,5}\left(\frac{29}{27}, \frac{1}{16}\right) > 0.1317993568270.$$

Warning 1.3 (What is not proved or used). The argument does not construct globally defined scalar functions whose ordinary-LCM depths are $0, 1, \ldots, s$. It does not reverse the Hodge flag, transport literal n^e denominators through a change of local coordinate, or replace the nonzero-weight Tate trivializations by rational functions of the Hauptmodul. The small-prime saving is only one power of the auxiliary prime; the large-prime theorem is proved directly on the de Rham frame torsor.

The paper is organized as follows. Section 2 states the published interfaces and fixes notation. Section 3 constructs the algebraic Eisenstein–Eichler packet and proves its raw arithmetic, BGG realization, functional rank, and zero estimate. Section 4 constructs the global source and proves the fixed-data descent and single-action projection bridges. The small-prime Hasse kernel and large-prime torsor Cartier theorem are developed in Section 5. Section 6 assembles these local estimates by Bost's slopes inequality. Analytic continuation and the exact collision energy are proved in Sections 7 and 8; the twenty-two applications and the interval certificate appear in Sections A and 9.

2 Published inputs and notation

We use the following published results.

(E1) **Overconvergent Eisenstein families.** Write $s = 2n+1$ and set $\kappa(a) = a^{-2n} = a^{1-s}$. In Calegari's notation,

$$\sigma^*_\kappa(m) = \sum_{\substack{d|m\\ p\nmid d}} \kappa(d)d^{-1} = \sum_{\substack{d|m\\ p\nmid d}} d^{-s}.$$

Thus Theorem 2.3 of [4] gives an overconvergent eigenform with the required nonconstant q-expansion, and Lemma 2.4 identifies its constant term as $\zeta_p(s)/2$. This is the form denoted $K_{p,s}$ below.

(E2) **Finite-slope continuation.** Buzzard's Theorem 5.2 in [3] extends an overconvergent modular form of integer weight and nonzero U_p-eigenvalue across the wide-open $W_0(p)$.

(E3) **Hasse invariants and q-expansions.** We use Katz's geometric Hasse invariant and the q-expansion principle from [8]. For every auxiliary prime $\ell \geq 5$, the classical congruences

$$E_{\ell-1} \equiv 1 \pmod{\ell}, \qquad E_{\ell+1} \equiv E_2 \pmod{\ell}$$

are used in q-expansions.

(E4) **The de Rham frame torsor.** Propositions A.14–A.15 of Graham–Pilloni–Rodrigues Jacinto [7] provide, at neat level, the algebraic Hodge-compatible B-torsor, its torus-weight decomposition, the filtration induced by the unipotent radical, and the Gauss–Manin connection. The descent from one fixed neat cover to the modular stack and coarse curve, and the uniform spreading out over $\mathbf{Z}[\Sigma^{-1}]$, are not quoted from that source; they are proved internally in Lemma 4.2.

(E5) **Arithmetic holonomy.** We use Bost's slopes inequality [1]; the arithmetic Hilbert–Samuel source-degree asymptotic together with the Bost–Charles overflow identities [2, Theorem 5.4.1 and Proposition 5.4.2], in the adelic assembly of [6, Theorem 2.6 and Section 2.3]; and the Chudnovsky–Osgood estimate and scalar source, pivot-height, and prime-window formulas of [5, Theorem 3.2.13 and Section 7.3, especially (7.3.7) and (7.3.14)–(7.3.16)]. The scalar local-denominator hypothesis is replaced by the hybrid finite-place theorem proved here; once the interval window is obtained, the CDT prime-number-theorem summation is used without alteration.

All structural algebraic objects independent of the hypothetical rational constant η are spread over $\mathbf{Z}[\Sigma^{-1}]$ after enlarging a fixed finite set Σ depending on p and s. Under the contradiction hypothesis $\eta \in \mathbf{Q}$, the finitely many primes dividing its denominator are treated separately; they contribute only $o(D^2)$. Such fixed changes alter neither a D^2 arithmetic-degree coefficient nor a leading pivot-height coefficient.

Let

$$f_p(\tau) = \left(\frac{\Delta(p\tau)}{\Delta(\tau)}\right)^{1/(p-1)}.$$

For $p \in \{2,3,5,7\}$ this is a Hauptmodul at the cusp $P = \infty$ and

$$f_p(q) = q \prod_{n\geq 1} \left(1 + q^n + \cdots + q^{(p-1)n}\right)^{24/(p-1)} \in q + q^2\mathbf{Z}[[q]]. \tag{6}$$

Its inverse satisfies

$$q(f_p) \in f_p + f_p^2\mathbf{Z}[[f_p]]. \tag{7}$$

We usually write $f = f_p$ and

$$D = q\frac{\mathrm{d}}{\mathrm{d}q}.$$

For an admissible adelic family of continuation templates $\boldsymbol{\phi} = (\phi_v)_v$, each centered at P, put

$$\Lambda(\boldsymbol{\phi}) = \sum_v \log|\phi_v'(0)|_v$$

and

$$\mathscr{H}(\boldsymbol{\phi}) = \sum_{v\mid\infty} \iint_{\mathbb{T}_v^2} \log|\phi_v(z) - \phi_v(w)|_v \,\mathrm{d}m(z)\,\mathrm{d}m(w) + \sum_{v\nmid\infty} \sup_{z\in\mathbb{T}_v} \log|\phi_v(z)|_v.$$

These are normalized as in [5, Section 7.3] and [6, Section 2].

3 The Eisenstein–Eichler packet

3.1 The negative-weight Eisenstein form

For odd $s \geq 3$ define

$$J_{p,s}(q) = \sum_{n\geq 1} \sigma_{-s}^{(p)}(n)q^n, \qquad \sigma_{-s}^{(p)}(n) = \sum_{\substack{d\mid n\\ p\nmid d}} d^{-s}, \tag{8}$$

and, for a scalar η,

$$K_{p,s,\eta} = J_{p,s} + \eta.$$

At

$$\eta = \zeta_p(s)/2,$$

input (E1) identifies $K_{p,s,\eta}$ as an overconvergent eigenform of weight $1 - s$.

Lemma 3.1 (U_p-eigenvalue)**.** *For every $n \geq 1$,*

$$\sigma_{-s}^{(p)}(pn) = \sigma_{-s}^{(p)}(n).$$

Consequently

$$U_p K_{p,s,\zeta_p(s)/2} = K_{p,s,\zeta_p(s)/2}.$$

Proof. The divisors of pn that are prime to p are precisely the divisors of n that are prime to p. The usual q-expansion definition of U_p now gives the assertion, including the constant term. □

3.2 Raw integration depths

For $1 \leq e \leq s$ put

$$R_e = D^{s-e} J_{p,s}. \tag{9}$$

Lemma 3.2 (Raw integration depth)**.** *For every $n \geq 1$,*

$$n^e [q^n] R_e = \sum_{\substack{d \mid n \\ p \nmid d}} (n/d)^s \in \mathbf{Z}. \tag{10}$$

Consequently, if $L_N = \operatorname{lcm}(1, \ldots, N)$, then

$$L_N^e R_e(q) \pmod{q^{N+1}}$$

has integral coefficients. The same is true after the integral substitution $q = q(f)$:

$$L_N^e R_e(q(f)) \pmod{f^{N+1}} \in \mathbf{Z}[[f]]/(f^{N+1}).$$

Proof. The first identity follows directly from $[q^n]R_e = n^{s-e}\sigma_{-s}^{(p)}(n)$. Since $n \mid L_N$ for $n \leq N$, the q-integrality follows. By (7), the coefficient of f^k after substitution uses only q^n with $n \leq k$, and all composition coefficients are integral. This proves the final assertion. □

Remark 3.3. The lemma transports a cumulative LCM cutoff. It does not assert the coordinatewise identity $n^e[f^n]R_e(f) \in \mathbf{Z}$, which need not be invariant under an integral tangent-to-identity change of parameter.

3.3 Exact divided Frobenius

Lemma 3.4 (Exact Dwork relation)**.** *Let $\ell \neq p$ be a rational prime. For $1 \leq e \leq s$,*

$$\boxed{R_e(q) - \ell^{-e} R_e(q^\ell) \in \mathbf{Z}_\ell[[q]].} \tag{11}$$

After $q = q(f)$,

$$R_e(q(f)) - \ell^{-e} R_e(q(\varphi_\ell(f))) \in \mathbf{Z}_\ell[[f]], \tag{12}$$

where

$$\varphi_\ell(f) = f(q(f)^\ell).$$

Proof. If $\ell \nmid n$, every divisor of n is an ℓ-adic unit. If $\ell \mid n$, the q^n-coefficient of the left side of (11) is
$$n^{s-e}\left(\sigma_{-s}^{(p)}(n) - \ell^{-s}\sigma_{-s}^{(p)}(n/\ell)\right).$$
The term in parentheses is
$$\sum_{\substack{d\mid n\\ p\nmid d\\ \ell\nmid d}} d^{-s},$$
which is ℓ-integral. Substitution of the integral inverse coordinate gives (12). $\square$

Lemma 3.5 (Coordinate Frobenius)**.** *For every prime ℓ outside a fixed exceptional set,*
$$\boxed{\varphi_\ell(f) = f^\ell + \ell\Delta_\ell(f), \qquad \Delta_\ell(f) \in f^{\ell+1}\mathbf{Z}_\ell[[f]].} \tag{13}$$

Proof. The product (6) is integral and begins with q. Hence in characteristic ℓ,
$$f(q)^\ell = f(q^\ell).$$
Substituting $q = q(f)$ gives $\varphi_\ell(f) \equiv f^\ell \pmod{\ell}$. Both sides have order ℓ and the same leading coefficient, so the quotient of their difference by ℓ has order at least $\ell+1$. $\square$

3.4 The split BGG extension

Let $\mathcal{H}$ be the rank-two de Rham bundle of the universal elliptic curve, with its logarithmic Gauss–Manin connection, and put
$$V = \operatorname{Sym}^{s-1}\mathcal{H}.$$
Let $L_{\mathrm{top}} \subset V$ be the top oper line and L_{bot} the bottom oper quotient. Locally choose an oper derivation ∂ and an oper frame $v_0, \dots, v_{s-1}$ such that
$$L_{\mathrm{top}} = \langle v_0\rangle, \qquad L_{\mathrm{bot}} = V/\langle v_0, \dots, v_{s-2}\rangle,$$
and
$$\nabla_\partial v_j = (s-1-j)v_{j+1}, \qquad v_s = 0.$$
For a local section F of $L_{\mathrm{bot}} \simeq \omega^{1-s}$, represented as a scalar in this oper gauge, define
$$\operatorname{spl}_s(F) = \sum_{j=0}^{s-1}(-1)^j\frac{(s-1)!}{(s-1-j)!}\partial^{s-1-j}F\,v_j. \tag{14}$$

Lemma 3.6 (Canonical BGG lift and normalization)**.** *The expression* (14) *is the unique local section G of V with the following two intrinsic properties:*

- (i) *the image of G in L_{bot} is $(-1)^{s-1}(s-1)!F = (s-1)!F$;*
- (ii) *$\nabla_V G$ takes values in $L_{\mathrm{top}} \otimes \Omega^1(\log D)$.*

Consequently the local expressions glue in the Zariski topology to a canonical algebraic differential operator
$$\operatorname{spl}_s : \omega^{1-s} \longrightarrow V$$
of order $s-1$. This is the GL_2 *dual-BGG lift in the present normalization; compare [14, Proposition 2.4.3]. Moreover,*
$$\nabla_V \operatorname{spl}_s(F) = \iota_s(\operatorname{Bol}_s F). \tag{15}$$
Here ι_s is normalized by the oper frame so that its image is the v_0-line, and $\operatorname{Bol}_s F = \partial^s F$ *in an oper coordinate. On the Tate disc this is D^sF, with $D = q\,\mathrm{d}/\mathrm{d}q$.*

Proof. Put $c_j = (-1)^j(s-1)!/(s-1-j)!$. The image of (14) in the bottom quotient is $c_{s-1}F = (s-1)!F$, because $s-1$ is even. For $j \geq 1$, the coefficient of $\partial^{s-j}Fv_j$ in $\nabla_\partial \operatorname{spl}_s(F)$ is

$$c_j + (s-j)c_{j-1} = 0.$$

Only the v_0-term $\partial^s F v_0$ remains, proving existence and (15) with the displayed normalization.

For uniqueness, let $W = \sum_{j=0}^{s-1} a_j v_j$ have zero image in L_{bot} and suppose $\nabla_V W$ lies in the top oper line. Then $a_{s-1} = 0$, and the vanishing of the v_j-coefficient of $\nabla_\partial W$ for $1 \leq j \leq s-1$ gives

$$\partial a_j + (s-j)a_{j-1} = 0.$$

Starting with $j = s-1$ and descending gives $a_{s-2} = \cdots = a_0 = 0$. Thus the two intrinsic properties determine the lift uniquely. The oper filtration, logarithmic connection, and recursion above are algebraic, so local formulas on overlapping algebraic oper charts have those same properties, agree, and glue to an algebraic differential operator over the field of definition.

For the exact Tate normalization, write

$$\delta_k^{[a]} = \delta_{k+2(a-1)} \cdots \delta_{k+2}\delta_k.$$

Urban's iterated Gauss–Manin formula [15, Proposition 2.4.1 and Remark 2.4.2] gives, in the polynomial realization on the de Rham torsor,

$$\delta_k^{[a]} F = \sum_{j=0}^{a} \binom{a}{j} \left(\prod_{b=1}^{j} (k+a-b) \right) X^j D^{a-j} F. \tag{16}$$

Although Urban writes the formula in the classical nonnegative-weight range, its proof is the formal induction from $\delta_k = D + kX$ and $[D, X] = -X^2$; both sides are polynomial in k. Hence (16) remains valid at the negative integer $k = 1-s$. Taking $a = s-1$, the product in the jth coefficient is $(-1)^j j!$, so (16) gives exactly the coefficients in (14), with no unrecorded scalar or sign. Taking instead $a = s$, every term with $j \geq 1$ contains the factor $k+s-1 = 0$, and hence

$$\delta_{1-s}^{[s]} F = D^s F.$$

The global derivation and weight decomposition on the Hodge-compatible frame torsor are those of [7, Propositions A.14–A.15]; thus this Tate formula is the torsor realization of the canonical lift just constructed. □

Put

$$E_{p,s+1}^{\mathrm{evil}} := D^s K_{p,s,\eta} = D^s J_{p,s}. \tag{17}$$

Its q-expansion is

$$\sum_{n \geq 1} \Big(\sigma_s(n) - \mathbf{1}_{p|n}\sigma_s(n/p) \Big) q^n,$$

so it is the algebraic p-stabilized Eisenstein form of weight $s+1$ and is independent of η.

Proposition 3.7 (Split algebraic Eisenstein–Eichler connection). *On the split vector bundle*

$$\mathcal{E}_{p,s} = \mathcal{O} \oplus V$$

define

$$\nabla_{\mathcal{E}}(a, v) = \Big(\mathrm{d}a, \nabla_V v - a\, \iota_s(E_{p,s+1}^{\mathrm{evil}}) \Big). \tag{18}$$

This is an algebraic regular-singular connection, and on the Tate formal disc

$$S_{p,s,\eta} = (1, \operatorname{spl}_s(K_{p,s,\eta}))$$

is horizontal.

Proof. The connection is algebraic because $E^{\mathrm{evil}}_{p,s+1}$ is algebraic; integrability is automatic on a curve and regular singularity follows from the logarithmic Gauss–Manin connection. By Lemma 3.6,

$$\nabla_V \operatorname{spl}_s(K_{p,s,\eta}) = \iota_s(E^{\mathrm{evil}}_{p,s+1}),$$

which is exactly the horizontality equation for the displayed vector. □

3.5 Rees–Bol jets

On the Tate de Rham torsor use Urban's polynomial coordinate X and operators, with the polynomial q-expansion normalization of [15, Proposition 2.4.1 and Remark 2.4.2] and the torsor formalism of [7],

$$\delta_k = D + kX, \qquad D = q\frac{\partial}{\partial q} - X^2\frac{\partial}{\partial X}, \qquad [D, X] = -X^2.$$

Define

$$\mathcal{K}_0 = K_{p,s,\eta}, \qquad \mathcal{K}_j = \delta_{1-s+2(j-1)} \cdots \delta_{1-s} K_{p,s,\eta} \quad (1 \le j \le s).$$

Proposition 3.8 (Integral Rees–Bol identity)**.** *For $0 \le j \le s$ there are integers $c_{j,h}$ such that*

$$\mathcal{K}_j = \sum_{h=0}^{j} c_{j,h} X^h D^{j-h} K_{p,s,\eta}, \qquad c_{j,0} = 1. \tag{19}$$

The transition from $(K, DK, \ldots, D^{s-1}K)$ to $(\mathcal{K}_0, \ldots, \mathcal{K}_{s-1})$ is lower triangular over $\mathbf{Z}[X]$ with diagonal 1, and

$$\mathcal{K}_s = D^s K = E^{\mathrm{evil}}_{p,s+1}.$$

For $0 \le j \le s-1$, the coefficient of $X^j K$ in $\mathcal{K}_j$ is

$$\boxed{c_{j,j} = (-1)^j \frac{(s-1)!}{(s-1-j)!}.} \tag{20}$$

The terminal case $j = s$ is the separate Bol identity $\mathcal{K}_s = D^s K$ and has no $X^s K$ term.

Proof. Induct using $\delta_k = D + kX$ and $D(X^h) = -hX^{h+1}$. All coefficients remain integral and the coefficient of the highest derivative remains one. At the terminal step every X-term has a vanishing falling-factorial coefficient, which is Bol's identity. For the top X-coefficient write $c_j = c_{j,j}$. The next operator gives $c_j = (j-s)c_{j-1}$ with $c_0 = 1$, proving (20). □

3.6 Functional rank and the zero estimate

Proposition 3.9 (Functional rank)**.** *The complex monodromy orbit of the homogenized horizontal leaf $S_{p,s,\eta}$ spans a space of dimension $s+1$.*

Proof. The period-polynomial cocycle is not identically zero. Otherwise, subtracting a polynomial of degree at most $s-1$ from the Eichler primitive would produce a modular form of weight $1-s<0$ with moderate growth at every cusp, hence the zero form, contradicting $D^sK = E^{\mathrm{evil}}_{p,s+1} \neq 0$.

Let W be the span of the cocycle values in $\mathrm{Sym}^{s-1}(\mathbf{C}^2)$. The cocycle relation shows that W is $\Gamma_0(p)$-stable. Since $\Gamma_0(p)$ is Zariski dense in $\mathrm{SL}_2(\mathbf{C})$ and $\mathrm{Sym}^{s-1}(\mathbf{C}^2)$ is irreducible, W is the full s-dimensional representation. Adjoining the homogenizing constant gives dimension $s+1$. □

Lemma 3.10 (Functional independence)**.** *In any rational algebraic frame of $\mathcal{E}_{p,s}$, the coordinate functions of the horizontal leaf are linearly independent over $\mathbf{C}(f)$, and hence over $\mathbf{Q}(f)$.*

Proof. If a $\mathbf{C}(f)$-valued rational covector annihilates one branch, its single-valuedness implies that it annihilates every analytic continuation of that branch. At a generic basepoint it therefore annihilates the monodromy orbit, which spans the fibre by Proposition 3.9. The covector is zero. □

Corollary 3.11 (Bost pivot set)**.** *Let V_D be the set of nonzero jumps of the vanishing-order filtration of the degree-D evaluation map defined in Section 4. For every $\varepsilon > 0$,*

$$\#V_D = (s+1)D + O_{p,s}(1), \qquad V_D \subset [0, (s+1+\varepsilon)D + O_{p,s,\varepsilon}(1)]. \tag{21}$$

Proof. Put $m = s+1$. Choose one fixed rational algebraic frame and multiply by a fixed rational normalizer which is regular and nonzero at P. Since the bundle and normalizer are fixed, the degree-D source embeds into

$$\left(\mathbf{Q}[f]_{\le D+C}\right)^m$$

for one constant C, and its evaluation is obtained from m holonomic power series which are independent over $\mathbf{C}(f)$ by Lemma 3.10. The Chudnovsky–Osgood estimate in the form [5, Theorem 3.2.13], applied with polynomial degree $D+C$, therefore bounds the vanishing order of every nonzero source element by

$$(m+\varepsilon)(D+C+1) + C_\varepsilon = (m+\varepsilon)D + O_{p,s,\varepsilon}(1),$$

after replacing ε by a slightly smaller positive number if necessary. This proves the asserted containment of V_D.

The evaluation map is injective: a source in its kernel would give a $\mathbf{C}(f)$-linear relation among the coordinate functions. Since every nonzero graded quotient of the one-variable vanishing filtration is one-dimensional, the number of jumps is the source rank. By Lemma 4.1, this is $mD + O_{p,s}(1)$. □

4 Global sources and fixed-data bridges

Let

$$0 = \mathcal{F}^{s+1} \subset \mathcal{F}^s \subset \cdots \subset \mathcal{F}^0 = \mathcal{E}_{p,s}$$

be the cyclic oper filtration, with rank-one graded lines L_i. On the dual bundle put

$$\mathcal{A}_i = \mathrm{Ann}(\mathcal{F}^{i+1}) \subset \mathcal{E}_{p,s}^\vee, \qquad 0 \le i \le s.$$

Choose a fixed divisor B, disjoint from the Tate cusp P, clearing all fixed poles used below, and define

$$M_D = H^0\left(X_0(p), \mathcal{E}_{p,s}^\vee(DP+B)\right), \qquad M_{D,i} = H^0\left(X_0(p), \mathcal{A}_i(DP+B)\right). \tag{22}$$

Because the extension bundle is split as $\mathcal{E}_{p,s} = \mathcal{O} \oplus V$, put also

$$M_D^0 = H^0\left(X_0(p), \mathcal{O}(DP+B)\right), \qquad M_D^+ = H^0\left(X_0(p), V^\vee(DP+B)\right). \tag{23}$$

Thus $M_D = M_D^0 \oplus M_D^+$, and Riemann–Roch gives

$$\dim_{\mathbf{Q}} M_D^+ = sD + O_{p,s}(1). \tag{24}$$

After spreading out, the same direct-sum decomposition holds for every $\mathbf{Z}_\ell$-source lattice outside the fixed model exceptional set.

Lemma 4.1 (Exact graded source and saturation). *For $D \gg_{p,s} 1$,*

$$M_{D,i}/M_{D,i-1} \simeq H^0\left(X_0(p), L_i^\vee(DP+B)\right).$$

In particular,

$$\dim_{\mathbf{Q}} M_D = (s+1)D + O_{p,s}(1), \qquad \dim_{\mathbf{Q}}(M_{D,i}/M_{D,i-1}) = D + O_{p,s}(1).$$

After enlarging one fixed exceptional set Σ, the induced filtration of the $\mathbf{Z}_\ell$-source lattice is saturated for every $\ell \notin \Sigma$, uniformly in D.

Proof. Spread the fixed exact sequences $0 \to \mathcal{A}_{i-1} \to \mathcal{A}_i \to L_i^\vee \to 0$, the point P, and the divisor B over $R = \mathbf{Z}[\Sigma^{-1}]$. Relative Serre vanishing gives a single D_0 such that the first higher direct image vanishes for all $D \geq D_0$. Pushforward is then exact and compatible with base change, and the integral quotients are finite locally free. Riemann–Roch on $X_0(p) \simeq \mathbf{P}^1$ gives the dimensions. □

Lemma 4.2 (Fixed neat-level descent and uniform spreading out). *After enlarging B and a fixed finite set Σ, the following hold.*

(i) *The de Rham frame torsor and Gauss–Manin connection may be constructed on one fixed neat tame cover and descended, through their canonical descent data, to the modular stack; the associated bundle $\mathcal{O} \oplus \mathrm{Sym}^{s-1}\mathcal{H}$, its connection, canonical BGG lift, and contraction pairing descend to $X_0(p) \setminus |B|$.*

(ii) *For every $D \gg 1$ and every $\lambda \in M_D$, the normalized scalar pairing may be computed on a flat algebraic torsor chart and has the form*

$$f^D\langle \lambda, S\rangle = H_\lambda(f,u) + \sum_{e=1}^{s} Q_{e,\lambda}(f,u)R_e(f), \tag{25}$$

where u denotes fibre coordinates,

$$\mathrm{supp}_f\, Q_{e,\lambda} \subset [-C, D+C],$$

and the fibre-variable degree is bounded by a constant depending only on p, s.

(iii) *All these objects have models over $\mathbf{Z}[\Sigma^{-1}]$, with Σ independent of D. Fixed changes of model or of B contribute $o(D^2)$ to global arithmetic degrees and summed pivot heights.*

Proof. Choose a fixed neat tame level $N \geq 5$, prime to p, and a finite Galois cover $Y \to \mathscr{X}_0(p)$. The frame torsor and connection of [7, Propositions A.14–A.15] are functorial in the elliptic curve and level structure, so the two pullbacks to $Y \times_{\mathscr{X}_0(p)} Y$ agree and satisfy the cocycle condition. They descend to the modular stack. The generic stabilizer -1 acts trivially on $\mathrm{Sym}^{s-1}\mathcal{H}$ because $s-1$ is even. Put the finitely many elliptic points into B; the associated bundle, the canonical lift of Lemma 3.6, and the invariant pairing then descend to the coarse curve away from B.

Choose a fixed rational algebraic frame on that affine open, regular at P. Since f is a rational coordinate with a simple zero at P, multiplication by f^D cancels the variable pole at P. The remaining coordinates have f-support $[-C, D+C]$ and only fixed poles away from P. Pullback to the frame torsor and use the fixed representation $1 \oplus \mathrm{Sym}^{s-1}$ together with the fixed-order BGG/Rees–Bol differential operator. This gives (25) with fibre degree bounded solely in terms of p, s.

Finally, the finite collection of bundles, matrices, divisors, torsor charts, and morphisms spreads out after clearing finitely many denominators. The torsor chart may be chosen smooth, hence flat, over the resulting base. A fixed model change distorts norms on a source of rank $O(D)$ and on $O(D)$ one-dimensional pivot quotients, hence contributes at most $O(D\log D) = o(D^2)$. $\square$

Proposition 4.3 (Single unipotent action on the deepest multiplier)**.** *Write $s = 2r+1$ and $H = D\log f$. Fix the algebraic Hodge splitting σ used below, regular at P, and write $x = X/H$ for the normalized unipotent coordinate. There are a fixed polynomial $g(f) \in \mathbf{Z}[\Sigma^{-1}][f]$ with $g(0)$ a unit and a constant C such that, for every $D \gg 1$ and every $\lambda \in M_D^+$, the multiplier of the normalized undifferentiated row H^rK in $f^D\langle\lambda, S\rangle$ has the form*

$$q_{0,\lambda}(f,x) = \sum_{j=0}^{s-1} Q_{j,\lambda}(f)x^j, \tag{26}$$

linear in λ, with

$$g(f)Q_{j,\lambda}(f) \in \mathbf{Z}[\Sigma^{-1}][f], \qquad \deg(gQ_{j,\lambda}) \le D + C. \tag{27}$$

The same assertions hold after reduction at every $\ell \notin \Sigma$. Only one degree-$(s-1)$ unipotent matrix occurs in (26); in particular there is no doubled $2(s-1)$ fibre degree.

Proof. Put $n = s-1$ and choose the monomial basis $e_j = e_0^{n-j}e_1^j$ of $\operatorname{Sym}^n \mathcal{H}$. For

$$u(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

one has, in this basis,

$$\operatorname{Sym}^n(u(x))e_j = \sum_{h=0}^{j} \binom{j}{h} x^{j-h}e_h. \tag{28}$$

Thus every entry of $\operatorname{Sym}^n(u(x))$, its inverse, and their dual matrices is an integral polynomial in x of degree at most n.

Write the coordinates of $f^D\lambda$ in the dual frame induced by the fixed splitting σ as $P_{0,\lambda}(f), \dots, P_{n,\lambda}(f)$. These are functions on the base and contain no fibre coordinate. In the canonical BGG lift (14), the undifferentiated K occurs only in the bottom extremal component. The Tate oper frame and the σ-frame differ by a diagonal torus factor and one unipotent factor $u(x)$. Transport the V-valued BGG lift to the σ-frame and keep the source covector in the σ-dual frame. By equivariance of contraction,

$$\langle P, \rho(u(x))W\rangle = \langle \rho^\vee(u(x)^{-1})P, W\rangle, \qquad \rho = \operatorname{Sym}^n,$$

so either description uses one matrix from (28), not a product of two such matrices. More explicitly, let coeff_K denote projection onto the literal undifferentiated raw K-row and normalize the coefficient by H^{-r}. Then

$$\begin{aligned} q_{0,\lambda}(f,x) &= H^{-r}\operatorname{coeff}_K\left\langle f^D\lambda, \rho(u(x))\operatorname{spl}_s(K)\right\rangle \\ &= H^{-r}\operatorname{coeff}_K\left\langle \rho^\vee(u(x)^{-1})f^D\lambda, \operatorname{spl}_s(K)\right\rangle. \end{aligned} \tag{29}$$

The Rees–Bol identity is the raw-derivative coordinate formula for this same transported BGG section; it is not a second frame transport. Since the undifferentiated K occurs only in the bottom extremal component of (14), (29) is a fixed row of one matrix in (28) applied to the base-coordinate vector P_λ. Hence it has x-degree at most $s-1$.

The factor H^{-r} is independent of x, and it exactly removes the diagonal weight of the bottom component. The resulting coefficient is weight zero and therefore descends from the torsor to a rational function on the base. More generally the diagonal factor contributes no additional variable because

$$\mathrm{wt}(\mathcal{K}_j) = 1 - s + 2j, \qquad \mathrm{wt}(H) = 2, \qquad \mathrm{wt}(H^{r-j}\mathcal{K}_j) = 0.$$

Finally, as in Lemma 4.2, a fixed polynomial $g(f)$, nonzero at P, clears all fixed poles of the coordinates $P_{j,\lambda}$ and the fixed poles introduced by the normalization H^{-r}; multiplication by f^D then gives degree $D + O(1)$. This proves (26)–(27). The entire construction spreads out with the fixed torsor and representation, so the same formula holds after every reduction at $\ell \notin \Sigma$. □

Lemma 4.4 (Adelic fixed-bundle norm comparison)**.** *Let $\mathcal{V}$ be a fixed rank-m vector bundle on $\mathbf{P}^1_{\mathbf{Q}}$, equipped with fixed adelic models and metrics, and put*

$$N_D = H^0\left(\mathbf{P}^1, \mathcal{V}(DP + B)\right).$$

Suppose that the scalar contraction with a fixed horizontal leaf is injective and that its pivot set satisfies a linear zero estimate. Relative to m scalar polynomial sources of degree D, replacing the source by N_D, shifting the scalar degrees by fixed integers, multiplying by a fixed normalizer regular and nonzero at P, changing a fixed algebraic frame, or changing finitely many local models alters both the source degree and the sum of evaluation heights by $o(D^2)$. Consequently

$$\widehat{\deg} N_D = \frac{m}{2}\mathscr{H} D^2 + o(D^2), \tag{30}$$

and

$$\sum_{n \in V_D} \sum_v h_v(\psi_D^{(n)}) \le \left(-\frac{m^2}{2}\Lambda + m\mathscr{H}\right) D^2 + \mathscr{A}_D + o(D^2). \tag{31}$$

Proof. By Birkhoff–Grothendieck there is a fixed global bundle isomorphism

$$\mathcal{V}(B) \simeq \bigoplus_{i=1}^m \mathcal{O}(a_i) \tag{32}$$

with fixed integers a_i. Transport the fixed adelic data through this isomorphism. At every place, the original and transported norms are uniformly equivalent on the fixed analytic template; outside a finite set of finite places the integral models agree up to units. Thus there are constants $C_v \ge 1$, equal to 1 for all but finitely many v, which bound the operator norms of the comparison and its inverse. If N_D^{spl} denotes the same source with the transported split norms, then

$$\left|\widehat{\deg} N_D - \widehat{\deg} N_D^{\mathrm{spl}}\right| \le \mathrm{rank}(N_D) \sum_v \log C_v = O(D).$$

This is the required determinant-norm comparison.

Under (32), the source is the direct sum of the scalar sources

$$H^0(\mathbf{P}^1, \mathcal{O}((D + a_i)P)), \qquad 1 \le i \le m,$$

up to fixed conventions for the divisor B. The scalar source-degree asymptotic, in the normalization of [5, (7.3.7)], and its adelic form in [6, Section 2.3 and the proof of Theorem 2.6], where the arithmetic Hilbert–Samuel theorem is combined with the Bost–Charles overflow identities [2, Theorem 5.4.1 and Proposition 5.4.2], gives

$$\frac{1}{2}\mathscr{H}(D + a_i)^2 = \frac{1}{2}\mathscr{H} D^2 + O(D)$$

for the ith summand. Summing and including the $O(D)$ norm distortion proves (30).

We next compare evaluation heights. In the split frame the contraction is a scalar Padé evaluation with m fixed holonomic coordinate functions and row degrees $D + a_i$. On each fixed continuation template the bundle comparison matrices are analytic and bounded; after the fixed normalizer has cleared the fixed poles, the same is true of the matrices needed for the local upper bound. At all but finitely many finite places these matrices are integral of norm one. The normalizer is regular and nonzero at P, so it preserves vanishing order, and its local sup norm contributes only a fixed factor. Consequently these comparisons contribute $O(1)$ per pivot; allowing the standard monomial/coefficient-lattice comparison gives the harmless bound $O(\log D)$ per pivot. There are $mD + O(1)$ pivots, so the total comparison error is $O(D \log D) = o(D^2)$.

Replacing $D + a_i$ by D changes the analytic $D\mathscr{H}$ term by $O(1)$ per pivot. It also shifts any finite-place multiplier window by only $O(1)$; the sum of the corresponding fixed boundary contributions is $O(D \log D)$. The scalar CDT estimates [5, (7.3.14)–(7.3.15)] therefore apply with unchanged D^2 coefficients. Finally, the $mD + O(1)$ pivot orders are distinct and nonnegative, whence

$$\sum_{n \in V_D} n \geq \frac{1}{2} m^2 D^2 + o(D^2).$$

Inserting this in the scalar height estimate proves (31). □

5 The hybrid local arithmetic theorem

Throughout this section, p and the odd integer s are fixed, and $\zeta_p(s) \in \mathbf{Q}$ is assumed. Put

$$\eta = \zeta_p(s)/2 \in \mathbf{Q}.$$

Let $\Sigma_{p,s}$ be the fixed geometric/model exceptional set, independent of η, enlarged to contain p and every prime $\ell \leq \max\{3, s\}$. Write $\eta = u/v$ in lowest terms, with $u \in \mathbf{Z}$, $v \in \mathbf{Z}_{>0}$, and put

$$\Sigma_\eta = \{\ell : \ell \mid v\}, \qquad t_\ell = v_\ell(v).$$

We call a prime ℓ *geometrically good* if $\ell \notin \Sigma_{p,s}$, and *arithmetically good* if it is geometrically good and $\ell \notin \Sigma_\eta$. Thus every geometrically good prime automatically satisfies $\ell > \max\{3, s\}$. The Hasse codimension theorem below is uniform at every geometrically good prime and is independent of η; the one-power saving is used only at arithmetically good primes. The explicit baseline lattice at the finite set $\Sigma_{p,s} \cup \Sigma_\eta$ is defined after the shifted cutoff $N^\sharp$ is fixed in (45). Constants below are uniform in D and in every prime to which the corresponding statement applies.

5.1 Weight-zero normalization and the comparison coordinate

Write $s = 2r + 1$ and put

$$A_j = H^{r-j} D^j J, \qquad C_j = H^{r-j} \mathcal{K}_j, \qquad H = D \log f. \tag{33}$$

Since $H = 1 + O(q)$, its positive and negative fixed powers are integral formal units outside a fixed exceptional set. With $x = X/H$, Proposition 3.8 and $K = J + \eta$ give, for $0 \leq j \leq s - 1$,

$$\boxed{C_j = \sum_{h=0}^{j} c_{j,h} x^h A_{j-h} + c_{j,j} \eta H^r x^j.} \tag{34}$$

Indeed $D^{j-h}K = D^{j-h}J$ unless $h = j$, when the undifferentiated constant η remains. Thus, for a formal linear combination of these Rees–Bol iterates with multipliers P_j, if

$$q_k = \sum_{j=k}^{s-1} c_{j,j-k} x^{j-k} P_j \qquad (0 \le k \le s-1), \tag{35}$$

then

$$\boxed{\sum_{j=0}^{s-1} P_j C_j = q_0(A_0 + \eta H^r) + \sum_{k=1}^{s-1} q_k A_k.} \tag{36}$$

In particular, the rational constant term and the depth-s row have the same multiplier q_0. This is an identity in the polynomial realization of the BGG lift; it is not an assertion that the functions C_j form a rational scalar coordinate system for the global source. Actual global contractions are handled by Equation (25) and Proposition 4.3.

Choose an algebraic Hodge splitting σ on an affine open containing the Tate cusp. Evaluating Urban's operator on a classical weight-k modular form gives $D_q + kX_\sigma$, whereas the Serre derivative is $D_q - kE_2/12$. Their difference is a meromorphic modular form of weight two. Thus

$$X_\sigma = -\frac{E_2}{12} + g_\sigma, \qquad g_\sigma \in M_2^{\mathrm{mer}}(\Gamma_0(p)), \tag{37}$$

and hence

$$x_\sigma = -\frac{E_2}{12H} + r_\sigma(f), \qquad r_\sigma(f) \in \mathbf{Q}(f). \tag{38}$$

Lemma 5.1 (Hasse algebraization and degree)**.** *For every geometrically good auxiliary prime ℓ, the reduction of x_σ belongs to $\mathbf{F}_\ell(f)$. Writing*

$$\bar{x}_\sigma = \frac{a_\ell(f)}{b_\ell(f)}$$

in lowest terms with $b_\ell(0) \neq 0$, one has

$$\delta_\ell := \max\{\deg a_\ell, \deg b_\ell\} \le c_p \ell + C, \qquad c_p = \frac{p+1}{12}. \tag{39}$$

Proof. Modulo ℓ, the Hasse invariant is represented by $E_{\ell-1}$, whose q-expansion is 1, and $E_{\ell+1} \equiv E_2$. Hence the reduction of (38) is represented by

$$-\frac{E_{\ell+1}}{12HE_{\ell-1}} + \bar{r}_\sigma(f),$$

a modular function on $X_0(p)_{\mathbf{F}_\ell}$. The numerator and denominator of the first term are sections of the same weight-$(\ell+1)$ line bundle. The classical weighted valence formula for $\Gamma_0(p)$ gives total orbifold zero degree

$$\frac{[\mathrm{SL}_2(\mathbf{Z}) : \Gamma_0(p)](\ell+1)}{12} = \frac{(p+1)(\ell+1)}{12}$$

for either section; see [8]. We use only this upper bound and its leading coefficient. Since the numerator and denominator have the same weight, their quotient is weight zero. On passage to the coarse genus-zero curve, the only convention change occurs at the finitely many elliptic points; the correction to the zero or pole order is bounded independently of ℓ. Adding the fixed rational term $\bar{r}_\sigma(f)$ changes numerator and denominator degrees by another $O_{p,s}(1)$. Hence

$$\max\{\deg a_\ell, \deg b_\ell\} \le \frac{p+1}{12}\ell + O_{p,s}(1),$$

which is (39). Regularity of the chosen splitting at the cusp and $H(0) = 1$ give $b_\ell(0) \neq 0$. □

5.2 The genuine-source Hasse kernel

The small-prime argument is made on the actual global positive source M_D^+, not on a putative scalar Rees–Bol coordinate block on the base. For $\lambda \in M_D^+$ let $q_{0,\lambda}(f,x)$ be the multiplier of the normalized deepest row H^rK from Proposition 4.3; after separating $K = J + \eta$ it multiplies $A_0 + \eta H^r$. On the splitting σ it is evaluated at $x = x_\sigma(f)$.

Theorem 5.2 (Uniform genuine-source Hasse codimension)**.** *There are constants D_0, C, depending only on p, s and the fixed models, such that the following holds. Let $D \geq D_0$ and let ℓ be geometrically good. Write*

$$\bar{x}_\sigma = \frac{a_\ell(f)}{b_\ell(f)}, \qquad \delta_\ell = \max\{\deg a_\ell, \deg b_\ell\}$$

as in Lemma 5.1. Then the reduction of the genuine positive source contains a subspace $\overline{K}_{D,\ell} \subset \overline{M}_D^+$ with

$$\boxed{\dim_{\mathbf{F}_\ell} \overline{K}_{D,\ell} \geq (s-1)\max\{0, D - \delta_\ell - C\} \geq (s-1)\max\{0, D - \lceil c_p \ell \rceil - C\},} \tag{40}$$

such that

$$q_{0,\bar{\lambda}}\left(f, \frac{a_\ell}{b_\ell}\right) = 0 \quad \text{in } \mathbf{F}_\ell[[f]] \qquad (\bar{\lambda} \in \overline{K}_{D,\ell}). \tag{41}$$

Any subspace of $\overline{K}_{D,\ell}$ lifts to a saturated direct summand $K_{D,\ell} \subset M_{D,\ell}^+$ satisfying

$$q_{0,\lambda}(f, x_\sigma(f)) \in \ell \mathbf{Z}_\ell[[f]] \qquad (\lambda \in K_{D,\ell}). \tag{42}$$

The constants, the geometric exceptional set, and this codimension statement are independent of η.

Proof. By Proposition 4.3, after multiplication by the fixed polynomial $g(f)$ with $g(0)$ a unit, one has

$$q_{0,\lambda}(f,x) = \sum_{j=0}^{s-1} Q_{j,\lambda}(f)x^j, \qquad \deg(gQ_{j,\lambda}) \leq D + C_0.$$

Reduction modulo ℓ therefore defines an $\mathbf{F}_\ell$-linear map on the actual global source

$$\begin{aligned} T_{D,\ell} : \overline{M}_D^+ &\longrightarrow \mathbf{F}_\ell[f]_{\leq D+(s-1)\delta_\ell + C_1}, \\ \bar{\lambda} &\longmapsto \bar{g}(f) b_\ell(f)^{s-1} q_{0,\bar{\lambda}}\left(f, \frac{a_\ell(f)}{b_\ell(f)}\right). \end{aligned} \tag{43}$$

Indeed, the jth summand after clearing b_ℓ^{s-1} has degree at most $D + C_0 + j\delta_\ell + (s-1-j)\delta_\ell$.

By (24) and base change,

$$\dim_{\mathbf{F}_\ell} \overline{M}_D^+ = sD + O_{p,s}(1),$$

whereas the target of (43) has dimension at most $D + (s-1)\delta_\ell + O_{p,s}(1)$. Rank–nullity gives

$$\dim \ker T_{D,\ell} \geq (s-1)(D - \delta_\ell) - C_2.$$

After increasing one fixed constant C, this implies the first lower bound in (40); Lemma 5.1 gives the second. Take $\overline{K}_{D,\ell}$ to be a subspace of the kernel of the displayed conservative dimension.

Since $g(0)$ and $b_\ell(0)$ are units, both are invertible in $\mathbf{F}_\ell[[f]]$. The identity $T_{D,\ell}(\bar{\lambda}) = 0$ therefore implies (41); equivalently, use that $\mathbf{F}_\ell[[f]]$ is a domain. Extend a basis of $\overline{K}_{D,\ell}$ to an $\mathbf{F}_\ell$-basis of $\overline{M}_D^+$ and lift the entire basis to the free lattice $M_{D,\ell}^+$. The lifted first vectors span a saturated direct summand $K_{D,\ell}$. The integral models of Lemma 4.2 make the multiplier integral and its reduction compatible with this lift. Its reduction is zero, proving (42). ☐

Remark 5.3. The proof uses only the actual global source, the degree-$(s-1)$ unipotent representation, and Hasse algebraization of x_σ. It does not identify the normalized Rees–Bol rows with scalar coordinates in a rational frame on the base curve.

Lemma 5.4 (Actual deepest-row decomposition)**.** *On the fixed splitting chart, for every $D \gg 1$ and every $\lambda \in M_D^+$ one may write the normalized genuine contraction as*

$$f^D\langle\lambda, S\rangle = G_\lambda(f,u) + q_{0,\lambda}(f,x_\sigma)H^r J + \eta q_{0,\lambda}(f,x_\sigma)H^r + \sum_{e=1}^{s-1} \widetilde{Q}_{e,\lambda}(f,u)R_e(f), \tag{44}$$

where G_λ is depth zero, the f-supports are contained in a fixed enlargement of $[-C, D+C]$, and all displayed multipliers are integral at every arithmetically good prime. Thus the multiplier of the unique depth-s row $R_s = J$ is the formal unit H^r times the genuine multiplier $q_{0,\lambda}$ of Proposition 4.3; every other nonalgebraic row has depth at most $s-1$.

Proof. Apply the genuine torsor decomposition (25) to the canonical BGG lift. By Proposition 4.3, the coefficient of its normalized undifferentiated component $H^rK = H^r(J+\eta)$ is $q_{0,\lambda}$. Separating $K = J + \eta$ gives the two middle terms of (44). All remaining components involve $D^jJ = R_{s-j}$ with $j \geq 1$, hence have depth at most $s-1$. The support and integrality assertions follow from the spread-out model in Lemma 4.2; fixed powers of H are integral formal units at an arithmetically good prime. □

5.3 The small-prime hybrid lattice

Fix $1 < \xi < s+1$. Choose once and for all a constant C_{sh} at least as large as the absolute value of every fixed negative f-support bound in Lemmas 4.2 and 5.4, and put

$$N_0 = \lfloor \xi D \rfloor, \qquad N^\sharp = N_0 + C_{\mathrm{sh}}, \qquad a_{\ell,D} = v_\ell(L_{N^\sharp}).$$

The fixed enlargement from N_0 to $N^\sharp$ ensures that a Laurent multiplier with lower support at least $-C_{\mathrm{sh}}$ can only use raw coefficients of index at most $N^\sharp$ in an evaluated coefficient through f^{N_0}. Since C_{sh} is fixed, $\log L_{N^\sharp} = \xi D + o(D)$. For every $\ell \in \Sigma_{p,s}$ choose a fixed integer $\beta_\ell \geq 0$ clearing the denominators of the chosen local model, frame, and fixed powers of H; put $\beta_\ell = 0$ for $\ell \notin \Sigma_{p,s}$. At every prime $\ell \in \Sigma_{p,s} \cup \Sigma_\eta$ with $\ell \leq N^\sharp$, define the baseline lattice

$$M_{D,\ell}^{\mathrm{bad}} = \ell^{\beta_\ell} M_{D,\ell}^0 \oplus \ell^{sa_{\ell,D}+\beta_\ell+t_\ell} M_{D,\ell}^+, \tag{45}$$

where $t_\ell = 0$ when $\ell \notin \Sigma_\eta$. The factor $\ell^{sa_{\ell,D}}$ clears every positive-depth raw row needed through f^{N_0} by Lemma 3.2; the fixed factor β_ℓ clears model and frame denominators on both split source blocks, and t_ℓ clears the denominator of η. Since the exceptional set is finite and $a_{\ell,D} = O_\ell(\log D)$, its total index contribution is $O_{p,s,\eta}(D \log D) = o(D^2)$. Omitting the one-power Hasse saving at these fixed primes costs only $O_{p,s,\eta}(D)$.

For each arithmetically good prime $\ell \leq N^\sharp$, use Theorem 5.2 to choose a saturated direct summand $K_{D,\ell} \subset M_{D,\ell}^+$ of conservative rank

$$r_{D,\ell}^* = (s-1)\max\{0, D - \lceil c_p \ell \rceil - C\}, \tag{46}$$

and choose a complement $K_{D,\ell}^c$ in $M_{D,\ell}^+$. With the exact split source $M_{D,\ell} = M_{D,\ell}^0 \oplus M_{D,\ell}^+$, define

$$M_{D,\ell}^{\mathrm{hyb}} = M_{D,\ell}^0 \oplus \ell^{sa_{\ell,D}-1} K_{D,\ell} \oplus \ell^{sa_{\ell,D}} K_{D,\ell}^c. \tag{47}$$

Lemma 5.5 (Small-prime integrality)**.** *Every unit vector of (47) has integral evaluation through f^{N_0}.*

Proof. Write $a = a_{\ell,D} \geq 1$. For $\lambda \in K_{D,\ell}$ use the actual genuine-source decomposition (44). By Theorem 5.2, $q_{0,\lambda}(f, x_\sigma(f)) \in \ell\mathbf{Z}_\ell[[f]]$, and H^r is a formal unit. Hence the depth-s row $J = R_s$ acquires the missing one power of ℓ, so $(sa-1)+1 = sa$ clears its $L^s_{N^\sharp}$ denominator. The constant term is integral because $\eta \in \mathbf{Z}_\ell$ at an arithmetically good prime and is multiplied by the same divisible coefficient.

Every remaining nonalgebraic row in (44) has depth $d \leq s-1$, while

$$sa - 1 \geq da$$

for $a \geq 1$. Its multiplier is integral by Lemma 5.4. Thus ℓ^{sa-1} clears every row on $K_{D,\ell}$; the complement is scaled by the worst depth sa. Apply Lemma 3.2 with cutoff $N^\sharp$ after the integral substitution $q = q(f)$. □

Proposition 5.6 (Small-prime determinant cost)**.** *The local index exponent is*

$$v_\ell[M_{D,\ell} : M^{\mathrm{hyb}}_{D,\ell}] = s^2 D a_{\ell,D} - r^*_{D,\ell} + O_{p,s}(a_{\ell,D}). \tag{48}$$

After summing over arithmetically good $\ell \leq N^\sharp$, and using (45) at the finite exceptional set, the total small-prime index is

$$\sum_{\ell \leq N^\sharp} \log[M_{D,\ell} : M^{\mathrm{small}}_{D,\ell}] = S_{p,s}(\xi)D^2 + o(D^2), \tag{49}$$

where $M^{\mathrm{small}}_{D,\ell}$ denotes $M^{\mathrm{hyb}}_{D,\ell}$ at arithmetically good primes and $M^{\mathrm{bad}}_{D,\ell}$ at primes in $\Sigma_{p,s} \cup \Sigma_\eta$. These cases are exhaustive by construction. We define the adelic source lattice M^{hyb}_D by these local lattices for $\ell \leq N^\sharp$ and by the unmodified source lattice for $\ell > N^\sharp$. Here

$$S_{p,s}(\xi) = s^2\xi - (s-1)\int_0^\xi (1 - c_p x)_+ \,\mathrm{d}x.$$

In particular, if $c_p\xi < 1$, then

$$S_{p,s}(\xi) = s^2\xi - (s-1)\left(\xi - \frac{p+1}{24}\xi^2\right). \tag{50}$$

Proof. The positive source has rank $sD + O(1)$. Hence

$$(sa-1)r^* + sa(sD + O(1) - r^*) = s^2 Da - r^* + O(a),$$

which proves (48). Now

$$\sum_{\ell \leq N^\sharp} a_{\ell,D} \log \ell = \log L_{N^\sharp} = \xi D + o(D).$$

The prime number theorem and partial summation give

$$\sum_{\ell \leq N^\sharp} r^*_{D,\ell} \log \ell = (s-1)D^2 \int_0^\xi (1 - c_p x)_+ \,\mathrm{d}x + o(D^2).$$

Removing the finite set $\Sigma_{p,s} \cup \Sigma_\eta$ from the two prime sums changes their leading terms by at most $O_{p,s,\eta}(D \log D)$. The baseline lattices (45) and the $O(a)$ errors also contribute $O_{p,s,\eta}(D \log D) = o(D^2)$. □

Lemma 5.7 (Residual small-prime heights)**.** *After the cutoff, the total residual positive denominator penalty at primes $\ell \leq N^\sharp$, summed over all pivots, is*

$$O_{p,s,\xi}(D^{3/2} \log D) = o(D^2).$$

Proof. Every coefficient index occurring at a pivot is $O_s(D)$ by Corollary 3.11. A residual prime power can occur only if

$$\ell^{a_{\ell,D}+1} > N^\sharp, \qquad \ell^{a_{\ell,D}+1} = O_s(D).$$

Since $a_{\ell,D}+1 \geq 2$, this forces $\ell = O_s(\sqrt{D})$. Summing proper prime-power excesses gives $O_s(\sqrt{D}\log D)$ for one pivot and therefore $O_s(D^{3/2}\log D)$ over $(s+1)D + O(1)$ pivots. This includes the finite bad primes: the fixed factors $\beta_\ell + t_\ell$ in (45) clear their model and constant denominators, and only the same prime-power excess beyond $L_{N^\sharp}$ remains. The one omitted Hasse power at arithmetically good primes is exactly restored by the deepest-row divisibility and creates no additional residual term. □

5.4 Large primes on the de Rham torsor

At primes $\ell > N^\sharp$ use the unmodified source lattice. By Lemma 4.2, after pullback to a flat integral torsor chart the normalized scalar evaluation has the decomposition

$$F_\lambda(f) = H_\lambda(f,u) + \sum_{e=1}^{s} Q_{e,\lambda}(f,u)R_e(f), \tag{51}$$

with f-support $[-C, D+C]$ and fixed fibre degree. The value of F_λ is independent of the torsor point because it is the contraction of a source with a horizontal section.

Insert Lemma 3.4 into (51):

$$F_\lambda = H_\lambda^\flat + \sum_{e=1}^{s} \ell^{-e} Q_{e,\lambda}(f,u)R_e(\varphi_\ell(f)), \tag{52}$$

where $H_\lambda^\flat$ is integral. The fibre variables are coefficients; no Frobenius action on them is used.

Lemma 5.8 (Coefficient-ring Taylor no-backflow)**.** *Let A be any $\mathbf{F}_\ell$-algebra and*

$$\Gamma(f,z) = \sum_{j=0}^{d} f^j P_j(z), \qquad d < \ell, \qquad P_j(z) \in A[[z]].$$

Suppose $\Gamma(f, f^\ell) \in f^n A[[f]]$ and $\Delta(f) \in f^{\ell+1}A[[f]]$. Then for every $t \geq 0$,

$$\Delta(f)^t\, \partial_z^t \Gamma(f,z)\Big|_{z=f^\ell} \in f^{n+t}A[[f]]. \tag{53}$$

Proof. Write $P_j(z) = \sum_r p_{j,r} z^r$. Since $0 \leq j < \ell$, the exponent $j + \ell r$ uniquely determines (j,r); no cancellation between different pairs is possible, even if A has zero divisors. Thus a nonzero $p_{j,r}$ satisfies $j + \ell r \geq n$. Terms with $r < t$ vanish after t derivatives. For $r \geq t$, after differentiating t times and multiplying by Δ^t, the corresponding order is at least

$$j + \ell(r-t) + t(\ell+1) = j + \ell r + t \geq n + t.$$

□

Lemma 5.9 (Pivot-range truncation)**.** *Let $\ell > N^\sharp$ and let*

$$F_\lambda(f) = c_n f^n + O(f^{n+1}), \qquad c_n \neq 0,$$

be a Bost pivot. Put $\widetilde{F}_\lambda = f^C F_\lambda$ and $\widetilde{n} = n + C$, where the fixed C shifts all multiplier f-degrees into $[0, D+2C]$. For all sufficiently large D, every coefficient through order $f^{\widetilde{n}}$ in (52) is unchanged if each $R_e(z)$ is replaced by its truncation $R_{e,<\ell}(z)$ below z^ℓ. The coefficients of every $R_{e,<\ell}$ are ℓ-integral. Consequently, if $v_\ell(c_n) < 0$, then

$$1 \leq -v_\ell(c_n) \leq s. \tag{54}$$

Proof. By Corollary 3.11, $\widetilde{n} = O_{p,s}(D)$. Since $\ell > N^\sharp = \lfloor \xi D \rfloor + C_{\mathrm{sh}}$ with $\xi > 1$, one has $D + 2C < \ell$ and $\widetilde{n} < \ell^2$ for all sufficiently large D. Consider a monomial $f^j z^r$ in a multiplier times $R_e(z)$. For a nonzero tth Taylor term one has $t \le r$, and at $z = f^\ell + \ell\Delta_\ell(f)$ its f-order is at least

$$j + \ell(r - t) + t(\ell + 1) = j + \ell r + t.$$

If $r \ge \ell$, this is at least ℓ^2, and hence is strictly beyond the pivot order. Thus terms of z-degree at least ℓ cannot affect c_n. For $r < \ell$, the index r is an ℓ-adic unit, so the raw-depth formula makes the coefficient of z^r in $R_e(z)$ ℓ-integral. After truncation the only negative power of ℓ in (52) is therefore the explicit factor ℓ^{-e}, with $1 \le e \le s$. This proves (54). $\square$

Let A_ℓ be the integral coefficient algebra of the chosen flat torsor chart and put $A = A_\ell/\ell A_\ell$, a nonzero unital $\mathbf{F}_\ell$-algebra. Let a Bost pivot have the displayed expansion. If $v_\ell(c_n) \ge 0$ there is no positive denominator penalty. Otherwise put

$$a = -v_\ell(c_n) > 0.$$

By Lemma 5.9, $1 \le a \le s$. Put

$$\widetilde{F}_\lambda = f^C F_\lambda, \qquad \widetilde{n} = n + C.$$

For $1 \le e \le s$ define the shifted truncated product

$$B_e(f, u, z) = f^C Q_{e,\lambda}(f, u) R_{e,<\ell}(z) \in A_\ell[f, z]. \tag{55}$$

By Lemma 5.9,

$$\deg_f B_e \le D + 2C < \ell, \qquad \deg_z B_e < \ell, \qquad B_e(f, u, 0) = 0.$$

Choose coefficientwise representatives $B_{e,0}, \dots, B_{e,e-1} \in A_\ell[f, z]$ satisfying

$$B_e \equiv \sum_{\nu=0}^{e-1} \ell^\nu B_{e,\nu} \pmod{\ell^e A_\ell[f, z]}, \tag{56}$$

with the same degree bounds and with $B_{e,\nu}(f, u, 0) = 0$. Such digits are obtained recursively coefficient by coefficient; flatness makes multiplication by ℓ injective in A_ℓ.

For $1 \le b \le s$ define in the special-fibre coefficient algebra

$$\Gamma_b(f, z) = \sum_{e \ge b} \overline{B}_{e,e-b}(f, u, z). \tag{57}$$

Let $\Pi_b(f)$ be the pole-grade-b associated-graded symbol of the shifted truncated expression

$$f^C H_\lambda^\flat + \sum_{e=1}^{s} \ell^{-e} B_e(f, u, \varphi_\ell(f)).$$

Taylor expansion at $\varphi_\ell(f) = f^\ell + \ell\Delta_\ell(f)$ gives the exact identity

$$\boxed{\Pi_b(f) = \sum_{t=0}^{s-b} \frac{\overline{\Delta}_\ell(f)^t}{t!}\, \partial_z^t \Gamma_{b+t}(f, z)\Big|_{z=f^\ell}.} \tag{58}$$

Indeed a digit $B_{e,\nu}$ in the tth Taylor term carries the factor $\ell^{\nu+t-e}$ and hence has pole grade $b = e - \nu - t$; grouping all terms of fixed b gives (58). Since $\ell > s$, every $t!$ occurring here is a unit. Notice that digitizing the full product B_e, rather than $Q_{e,\lambda}$ alone, also records positive ℓ-adic valuations of the integral coefficients of $R_{e,<\ell}$.

Proposition 5.10 (Torsor Cartier strictness)**.** *For the actual pole grade a, the coefficient of $f^{\widetilde{n}}$ in $\Gamma_a(f, f^\ell)$ is nonzero. Consequently there exist*

$$-C \le j \le D + C, \qquad r \ge 1,$$

such that

$$n = j + \ell r. \tag{59}$$

Hence the positive denominator penalty satisfies

$$\delta_\ell(\psi_D^{(n)}) \le s\, v_\ell\left([\max\{n - D - C, 1\}, \dots, n + C]\right) \log \ell. \tag{60}$$

Proof. The shifted scalar series has expansion

$$\widetilde{F}_\lambda = c_n f^{\widetilde{n}} + O(f^{\widetilde{n}+1}).$$

For every $k < \widetilde{n}$ its f^k-coefficient is zero. Taking successive positive pole-grade symbols of that zero coefficient shows

$$\Pi_b(f) \in f^{\widetilde{n}} A[[f]] \qquad (1 \le b \le s).$$

We prove by descending induction on b that

$$\Gamma_b(f, f^\ell) \in f^{\widetilde{n}} A[[f]], \qquad \Pi_b(f) \equiv \Gamma_b(f, f^\ell) \pmod{f^{\widetilde{n}+1}}. \tag{61}$$

For $b = s$, (58) has only the term $t = 0$, so the assertion follows. Suppose it is known for $b+1, \dots, s$. For every $t \ge 1$, the induction hypothesis gives $\Gamma_{b+t}(f, f^\ell) \in f^{\widetilde{n}} A[[f]]$. Since $\deg_f \Gamma_{b+t} < \ell$, Lemma 5.8 yields

$$\frac{\overline{\Delta}_\ell(f)^t}{t!}\, \partial_z^t \Gamma_{b+t}(f, z)\Big|_{z=f^\ell} \in f^{\widetilde{n}+t} A[[f]] \subset f^{\widetilde{n}+1} A[[f]].$$

The $t = 0$ term is $\Gamma_b(f, f^\ell)$, proving the congruence in (61); because $\Pi_b \in f^{\widetilde{n}} A[[f]]$, it also proves the asserted divisibility of Γ_b.

At the actual pole grade a, the coefficient of $f^{\widetilde{n}}$ in Π_a is the reduction of the unit $\ell^a c_n$. It is a nonzero scalar in $\mathbf{F}_\ell$ and remains nonzero in the nonzero unital algebra A. By (61), the coefficient of $f^{\widetilde{n}}$ in $\Gamma_a(f, f^\ell)$ is therefore nonzero. Some monomial contributing to it has exponent

$$\widetilde{n} = j' + \ell r, \qquad 0 \le j' \le D + 2C.$$

Every digit $B_{e,\nu}$ has zero z-constant term, so $r \ge 1$. Returning to the unshifted order and putting $j = j' - C$ gives

$$n = j + \ell r, \qquad -C \le j \le D + C.$$

The positive integer $n - j = \ell r$ lies in $[\max\{n - D - C, 1\}, n + C]$, so the ℓ-adic valuation of the lcm of this interval is at least one. Since $a \le s$, this proves (60). ☐

Remark 5.11 (Signed height versus positive penalty). Bost's local height is the signed quantity $h_\ell^B = -v_\ell(c_n) \log \ell$. The local Dwork estimates bound $\delta_\ell = \max\{0, -v_\ell(c_n)\} \log \ell$, and $h_\ell^B \le \delta_\ell$. Replacing the signed height by this upper bound is therefore legitimate.

5.5 The aggregate large-prime sum

For $m \geq 2$ and $1 < \xi < m$, define $I_\xi^m(\xi)$ by (1).

Lemma 5.12 (CDT prime-window summation)**.** *If V_D satisfies (21), then*

$$\sum_{n \in V_D} \sum_{\ell > \xi D} v_\ell([\max\{n-D, 1\}, \ldots, n]) \log \ell \leq I_\xi^m(\xi) D^2 + o(D^2).$$

Proof. This is the one-variable summation in [5, Section 7.3]. A prime $\ell = xD$ can contribute only if a pivot lies within one multiplier window of a positive multiple of ℓ. The distinct pivots are maximally packed in an interval of normalized length m. Integrating the resulting packing function and partitioning at the points $1 + k\xi$ gives (1). □

Proposition 5.13 (Large-prime arithmetic cost)**.** *The total residual positive denominator penalty at primes $\ell > N^\sharp$ is*

$$\mathscr{A}_D^{\text{large}} \leq s I_\xi^{s+1}(\xi) D^2 + o(D^2).$$

Proof. Since $N^\sharp = \xi D + O(1)$, the sum over $\ell > N^\sharp$ is bounded by the CDT sum over $\ell > \xi D$, up to a fixed-width boundary error. Apply Proposition 5.10 and Lemma 5.12. Adding and removing the fixed C-width changes the prime sum by $O_{p,s}(D \log D)$, because only fixed boundary integers are added for each of $O_s(D)$ pivots. □

Theorem 5.14 (Hybrid finite-place theorem)**.** *For the hybrid source lattice, the finite-place contributions satisfy*

$$\widehat{\deg} M_D^{\text{hyb}} = \widehat{\deg} M_D - S_{p,s}(\xi) D^2 + o(D^2),$$
$$\mathscr{A}_D^{\text{small}} = o(D^2), \qquad \mathscr{A}_D^{\text{large}} \leq s I_\xi^{s+1}(\xi) D^2 + o(D^2).$$

Proof. Combine Propositions 5.6 and 5.13 and Lemma 5.7. □

6 Bost-slopes assembly

Proposition 6.1 (Bost–CDT leading terms)**.** *For the unmodified rank-m source,*

$$\widehat{\deg} M_D = \frac{m}{2} \mathscr{H}(\phi) D^2 + o(D^2). \tag{62}$$

For the Bost pivots, the nonarithmetic template terms satisfy

$$\sum_{n \in V_D} \sum_v h_v(\psi_D^{(n)}) \leq \left(-\frac{m^2}{2} \Lambda(\phi) + m \mathscr{H}(\phi) \right) D^2 + \mathscr{A}_D + o(D^2). \tag{63}$$

Proof. For the split scalar source, the scalar source-degree asymptotic is [5, (7.3.7)]. In the adelic normalization, [6, Section 2.3 and the proof of Theorem 2.6] derives the same leading term from the arithmetic Hilbert–Samuel theorem together with the Bost–Charles overflow identities [2, Theorem 5.4.1 and Proposition 5.4.2]. The fixed-bundle comparison Lemma 4.4 therefore gives (62).

For a pivot of order n, the constituent CDT estimates have nonarithmetic part

$$-n\Lambda(\phi) + D\mathscr{H}(\phi),$$

while the finite coefficient-arithmetic term is the quantity denoted $\mathscr{A}_D$ here; see [5, (7.3.14)–(7.3.15)]. There are $mD + O(1)$ distinct nonnegative pivots and, by Corollary 3.11, each is $O(D)$. Hence

$$\sum_{n \in V_D} n \geq \binom{\# V_D}{2} = \frac{1}{2} m^2 D^2 + o(D^2),$$

and the uniform lower-order terms sum to $o(D^2)$. Summing the CDT estimates and retaining $\mathscr{A}_D$ separately gives (63); this is the leading-coefficient passage in [5, (7.3.14)–(7.3.16)], with the adelic normalization above. □

Proof of Theorem 1.2. Bost's slopes inequality, in the form of [5, (7.3.16)], gives

$$\widehat{\deg} M_D^{\mathrm{hyb}} \leq \sum_{n \in V_D} \sum_v h_v(\psi_D^{(n)}).$$

By Theorem 5.14 and Proposition 6.1,

$$\frac{m}{2}\mathscr{H} - S_{p,s}(\xi) \leq -\frac{m^2}{2}\Lambda + m\mathscr{H} + sI_\xi^{s+1}(\xi).$$

Rearranging and using $m = s + 1$ gives

$$m\left(\Lambda - \frac{2}{m^2}(S_{p,s}(\xi) + sI_\xi^{s+1}(\xi))\right) \leq \mathscr{H},$$

which is (3). □

7 Uniform analytic continuation

Put

$$a_p = \frac{12}{p-1}, \qquad R_p = p^{a_p}.$$

Theorem 7.1 (Uniform continuation)**.** *For $p \in \{2, 3, 5, 7\}$ and odd $s \geq 3$, the distinguished form $K_{p,s,\zeta_p(s)/2}$ extends as a rigid analytic section of ω^{1-s} to*

$$U_p = \{|f_p|_p < R_p\}.$$

The full Eisenstein–Eichler horizontal leaf extends meromorphically over the same wide-open. At every finite place $\ell \neq p$, under $\zeta_p(s) \in \mathbf{Q}$, it is meromorphic on $|f_p|_\ell < 1$. For every $Y > 0$, its complex pullback under $z \mapsto f_p(e^{-2\pi Y} z)$ is meromorphic near the closed unit disc.

Proof. Input (E1) makes K overconvergent and Lemma 3.1 gives U_p-eigenvalue one. The weight $1 - s$ is an integer, and Buzzard's theorem imposes no nonnegativity condition on that integer weight. Pull to one fixed neat tame level, apply [3, Theorem 5.2(2)], and descend by uniqueness of analytic continuation. Thus K extends to the wide-open $W_0(p)$ containing the cusp ∞.

For $p = 2, 3$, Calegari identifies the two ordinary components and the intervening supersingular region in the coordinate f_p, giving $W_0(p) = \{|f_p|_p < R_p\}$; see [4, Section 3]. We spell out the corresponding calculation for $p = 5, 7$. Put $t_p = f_p^{-1}$. For $p = 5$, Kilford's equation [9, (3)] is

$$j = \frac{(t_5^2 + 250t_5 + 3125)^3}{t_5^5}.$$

Writing $v = v_5(t_5)$, its Newton-polygon valuation gives

$$v_5(j) = v \quad (0 < v < 5/2), \qquad v_5(j) = 15 - 5v \quad (5/2 < v < 3).$$

Since $j = 0$ is the unique supersingular j-invariant in characteristic 5, the supersingular annulus is $0 < v_5(t_5) < 3$. The cusp ∞ lies on the side where t_5 has a pole; hence

$$W_0(5) = \{v_5(t_5) < 3\} = \{|f_5|_5 < 5^3\}.$$

For $p = 7$, Kilford–McMurdy give $w_7^* t_7 = 49/t_7$ and, from their explicit j-models, identify the unique supersingular annulus as $0 < v_7(t_7) < 2$; see [10, Equations (1)–(4) and Section 2]. Thus

$$W_0(7) = \{v_7(t_7) < 2\} = \{|f_7|_7 < 7^2\}.$$

The Gauss–Manin connection is algebraic on the de Rham frame torsor, so applying it finitely many times does not shrink the base domain. This extends the BGG leaf. At $\ell \neq p$, the raw coefficients have at worst logarithmic ℓ-adic denominator growth and hence radius at least one; the integral inverse coordinate transfers this to the f-disc. Over $\mathbf{C}$, the divisor-sum coefficients have subexponential growth, giving the final claim. □

Remark 7.2 (Strict exhaustion). The p-adic continuation domain is open. In the holonomy argument one uses closed affinoids of radius $p^{a_p - \varepsilon}$ and lets $\varepsilon \downarrow 0$. No boundary convergence is assumed.

8 Exact modular collision energy

For $0 < Y < 1/p$, define

$$\phi_{p,Y}(z) = f_p(e^{-2\pi Y} z)$$

and $C_p(Y)$ by (4).

Proposition 8.1 (Modular Jensen formula). *One has*

$$\iint_{\mathbb{T}^2} \log |\phi_{p,Y}(z) - \phi_{p,Y}(w)| \; \mathrm{d}m(z)\, \mathrm{d}m(w) = -2\pi Y + C_p(Y). \tag{64}$$

Proof. Fix $\tau_x = x + iY$. Since f_p is a Hauptmodul, the nonidentity zeros of $f_p(q') - f_p(e^{2\pi i \tau_x})$ are parametrized by cosets $\Gamma_\infty \backslash \Gamma_0(p)$, equivalently primitive bottom rows (c, d) with $c > 0$ and $p \mid c$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$\Im(\gamma \tau_x) = \frac{Y}{(cx + d)^2 + c^2 Y^2}.$$

Jensen's formula contributes $2\pi(\Im(\gamma\tau_x) - Y)_+$. For a fixed c, unfold the sum over primitive d:

$$\sum_{\substack{d \in \mathbf{Z} \\ (c,d)=1}} \int_0^1 F(cx + d)\; \mathrm{d}x = \frac{\varphi(c)}{c} \int_{\mathbf{R}} F(u)\; \mathrm{d}u.$$

The positive part is nonempty exactly when $cY < 1$. Evaluating the elementary integral gives

$$\frac{4\pi\varphi(c)}{c^2} \left(\arccos(cY) - cY\sqrt{1 - c^2 Y^2} \right).$$

Only finitely many c occur. The eta-product factors in f_p have logarithmic mean zero, while the leading q contributes $-2\pi Y$. Elliptic stabilizers occur only on a measure-zero set and are accounted for by the standard orbifold multiplicity. □

At the distinguished p-adic place, strict exhaustion of the radius R_p gives a scaling width $12 \log p/(p - 1)$. Combining it with Proposition 8.1 yields

$$\Lambda_p(Y) = \frac{12 \log p}{p - 1} - 2\pi Y, \qquad \mathscr{H}_p(Y) = \Lambda_p(Y) + C_p(Y). \tag{65}$$

9 The twenty-two applications

For each target we use exact rational witnesses ξ, Y listed in Table 1. Every witness satisfies

$$1 < \xi < s+1, \qquad c_p\xi < 1, \qquad 0 < Y < 1/p.$$

The value of $I_\xi^{s+1}(\xi)$ and hence $\tau_{p,s}(\xi)$ is rational. The collision sum has exactly the finitely many active layers $c = p, 2p, \ldots$ with $cY < 1$.

Table 1: Exact rational witnesses and independently certified lower margins.

p	s	ξ	$\tau_{p,s}(\xi)$	Y	certified lower bound
2	3	22/19	1321/608	16/79	9.745786633966
2	5	29/26	18445/5616	6/49	13.454727090366
2	7	110/101	105585/25856	1/11	15.805524331097
2	9	89/83	10906993/2324000	3/43	17.190005995811
2	11	262/247	464385991/89631360	3/52	17.790824360622
2	13	181/172	144989487/25958240	2/41	17.737379764768
2	15	478/457	16666007411/2810615808	3/71	17.124944575766
2	17	305/293	12175769323/1954839744	3/80	16.023242852853
2	19	758/731	11693630797/1801184000	1/30	14.484310655378
2	21	461/446	536638388131/79764338368	1/33	12.553794492186
2	23	1102/1069	22108077629089/3185347156992	1/36	10.264641765760
2	25	649/631	21299386717285/2985462031552	1/39	7.646580292916
2	27	1510/1471	456079886220493/62372115303680	1/42	4.724811329943
2	29	869/848	14321093408042087/1915475381820000	1/46	1.522247623229
3	3	33/29	2029/928	4/27	6.200294149415
3	5	87/79	112477/34128	7/73	6.941905556192
3	7	55/51	53435/13056	1/15	6.163732078651
3	9	267/251	33037241/7028000	1/19	4.332047506141
3	11	393/373	702154669/135354240	3/70	1.679644254174
5	3	11/10	177/80	7/71	2.515479614080
5	5	29/27	12899/3888	1/16	0.131799356827
7	3	33/31	2219/992	1/14	0.436413057785

Proof of Theorem 1.1. Fix one row of Table 1 and suppose $\zeta_p(s) \in \mathbf{Q}$. By Theorem 7.1 and Proposition 8.1, the templates have width and energy (65). Apply Theorem 1.2 with the displayed ξ. It forces

$$(s+1)(\Lambda_p(Y) - \tau_{p,s}(\xi)) \le \Lambda_p(Y) + C_p(Y),$$

or equivalently

$$\mathfrak{M}_{p,s}(\xi, Y) = s\Lambda_p(Y) - (s+1)\tau_{p,s}(\xi) - C_p(Y) \le 0.$$

The interval certificate gives a strictly positive lower bound in every row, a contradiction. □

10 Logical scope and dependency audit

The proof is intentionally organized so that every delicate interface can be checked independently.

Gate	Source	Use in the proof

Calegari Eisenstein family	[4, Theorem 2.3 and Lemma 2.4]	Identifies $K_{p,s}$ and gives overconvergence.
Buzzard continuation	[3, Theorem 5.2]	Extends the finite-slope form to $W_0(p)$.
Hasse invariant	[8] plus classical Eisenstein congruences	Algebraizes x_σ modulo an auxiliary prime and bounds its degree.
De Rham frame torsor	[7, Propositions A.14–A.15]	Supplies the neat-level torsor, filtration, weight decomposition, and Gauss–Manin structure. Descent and spreading out are internal.
Bost/CDT	[1]; [2, Theorem 5.4.1 and Proposition 5.4.2]; [5, Theorem 3.2.13 and (7.3.7), (7.3.14)–(7.3.16)]; [6, Theorem 2.6]	Supplies the slopes inequality, overflow/source identities, scalar zero estimate, pivot-height bounds, adelic normalization, and prime-window summation; Lemma 4.4 proves the fixed-bundle norm comparison.
BGG extension	[14, Proposition 2.4.3]; [15, Proposition 2.4.1 and Remark 2.4.2]; internal Lemma 3.6 and Proposition 3.7	Fixes the exact global BGG–Bol normalization and constructs the algebraic Eisenstein–Eichler connection.
Small-prime arithmetic	Internal, Propositions 4.3 and 5.6, Theorem 5.2, and Lemma 5.4	Proves the normalized deepest-row degree bound, genuine-source Hasse rank–nullity saving, fixed-width cutoff control, and determinant cost.
Large-prime arithmetic	Internal, Lemmas 5.8 and 5.9 and Proposition 5.10	Truncates before taking pole grades, digitizes each full truncated product, and gives the torsor-valued Cartier window without scalarization.
Collision energy	Internal, Proposition 8.1	Gives the exact complex Bost–Charles energy.
Numerics	Appendix A	Gives exact rational witnesses and outward interval enclosures.

The attribution boundary is as follows. GPRJ supplies the torsor and its Gauss–Manin structures at neat level; Lemma 4.2 proves the descent and integral spreading used here. CDT supplies scalar source and pivot estimates; Lemma 4.4 proves that passage to the fixed vector-bundle source changes only $o(D^2)$. Katz supplies the Hasse invariant and q-expansion principle; Proposition 4.3 proves the single-action degree bound for the genuine deepest multiplier, and the codimension estimate is the internal rank–nullity argument of Theorem 5.2.

Warning 10.1 (Obsolete proof routes). The manuscript does not inherit the global scalar-gauge assertion from older prime-window drafts. A nonzero-weight Tate trivialization transforms by a nontrivial power of $c\tau + d$ and therefore cannot equal a rational function of the Hauptmodul. The Rees–Bol matrix controls the unipotent coordinate, not the diagonal torus. This is why the large-prime proof stays on the frame torsor and the small-prime proof uses only a mod-ℓ Hasse comparison.

A Independent rational fixed-point interval certificate

The supplementary program `hybrid_rational_interval_certificate.py` reconstructs every numerical quantity independently from the formulas of the paper. It uses only Python's standard-library arbitrary-precision integers and `Fraction`; no binary floating point or external interval package enters the certification. All algebraic quantities are exact. Real intervals are represented by integer endpoints at the fixed scale 10^{90}, and every arithmetic operation is rounded outward.

The transcendental constants are enclosed from elementary convergent series with explicit remainder bounds:

$$\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}, \qquad \log p = 2 \operatorname{arctanh} \frac{p-1}{p+1},$$

$$\arctan z = \sum_{k \geq 0} \frac{(-1)^k z^{2k+1}}{2k+1}, \qquad \arccos x = 2 \arctan \sqrt{\frac{1-x}{1+x}}.$$

The alternating arctangent remainder is bounded by the first omitted term; the positive arctanh tail is bounded by a geometric series. Square roots of rationals are enclosed by exact integer square root at scale 10^{90}. Thus every printed interval follows from integer inequalities that can be checked independently of a transcendental-function library.

For each pair (p, s) the script computes

$$\xi = \frac{12(s(s+1)-1)}{12s^2 + (s-1)(p+1)},$$

then evaluates $K_m(\xi)$, $I_\xi^m(\xi)$, $S_{p,s}(\xi)$, and $\tau_{p,s}(\xi)$ exactly. The active collision layers are selected by the exact rational test $cY < 1$. The program asserts that the lower endpoint of every one of the twenty-two final intervals is positive. Its smallest case is

$$\mathfrak{M}_{5,5}\left(\frac{29}{27}, \frac{1}{16}\right) \in [0.131799356827016832557664457131479890735826671587851068262655,$$
$$0.131799356827016832557664457131479890735826671587851068262656].$$

For reproducibility, the accompanying manifest `hybrid_rational_interval_certificate_sha256.txt` records the SHA-256 hashes of the certificate script and its output:

```
1408c9092d0c34917253c8f520db56853112c58640d15c98d58b76a61a0478f5
  hybrid_rational_interval_certificate.py
9e1d8f74eb198c7e7b0c7420a1e8021d7bad4d2bc8014cc939b6b353111c0da0
  hybrid_rational_interval_certificate_output.txt
```

B Explicit prime-window function

For completeness, let $m \geq 2$, $1 < \xi < m$, and $K = \lfloor (m-1)/\xi \rfloor$. The normalized pivot interval has length m and a large prime $\ell = xD$ contributes when a pivot lies in a degree-D interval ending at a positive multiple of ℓ. Maximally packing distinct pivots and integrating the resulting triangular boundary term gives

$$I_\xi^m(\xi) = \left(m - \frac{1}{2}\right) H_K - K\xi + \frac{(m-(K+1)\xi)_+^2}{2(K+1)}.$$

This is the specialization of the CDT summation used in Lemma 5.12.

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